Lesson 5 — What Each Notebook Shows

Notebook guide, lesson 5 — Technologies for Artificial Intelligence

Fabio Antonini — Università degli Studi dell'Aquila

6 November 2026 · three notebooks

Contents

Notebook 01 — `01_why_one_split_lies` **1**
Why one train/test split lies 1

Notebook 02 — `02_bias_variance_and_learning_curves` (self-study on 6 November) **2**
What the error is made of 2

Notebook 03 — `03_leakage_and_honest_search` **4**
Leakage that survives cross-validation 4

A short guide to this lesson's notebooks. For each one: what it is for, the concepts in the order they appear, the numbers worth noticing, and **the two-minute version**, a summary to read before running it or to come back to afterwards. Every number here comes from the committed output of the notebook it belongs to.

Notebook 01 — `01_why_one_split_lies`

Why one train/test split lies

What it is for. Lesson 1 gave the rule: hold data out, never look at it, report the score. This notebook shows what the rule does **not** give you — the precision of the number it produces.

The data. Lesson 4's drives, cut down to **800 with 29 failures (3.6%)** — because 800 rows with 29 positives is the ordinary case, not an exaggerated one.

The concepts, in order.

1. **Two honest experiments, two answers.** Five seeds give test AUCs of 0.961, 0.966, 0.980, **1.000** and 0.990, each with 7 failures in the test set. Everything done correctly; only the rows held out changed.
2. **How wide is it?** 200 splits: worst **0.885**, best **1.000**, spread **0.115**, mean 0.955 with a standard deviation of 0.024, and **10 of 200 land above 0.99**. The right-hand tail is the dangerous end — a disappointing score makes you keep working, a delightful one makes you stop and publish.

3. **Choosing with a noisy number.** Three logistic regressions differing only in penalty: mean area under the receiver operating characteristic curve (AUC) **0.9549, 0.9549, 0.9544** — identical to three decimals — and they win outright **89, 42 and 38 times** out of 200, with 31 ties, by a median margin of **0.0015** against the 0.024 a single measurement wobbles by. The winner is noise wearing the clothes of a result.

4. **k-fold cross-validation**, implemented by hand and matched against scikit-learn **exactly** (agreement 0.00e+00), giving a mean of **0.9514** and a standard deviation of **0.0187**. The five rounds fit five different models — power-on hours weighs 0.010 in one and 0.353 in another — so the mean belongs to the procedure. And 300 new fleets show the real spread of the cross-validated number, **0.0165**, about twice the 0.0084 that s ÷ √5 promises.

5. **Stratification is not optional here.** Plain `KFold` gives folds with 1, 5, 7, 7 and 9 failures out of 160 — and the fold with one failure scores a perfect **1.000** — while `StratifiedKFold` gives 5, 6, 6, 6, 6 against a fleet-wide rate of 3.62%.

6. **How much it helps**, the comparison across 40 seeds:

	single 75/25 split	5-fold CV
worst	0.9119	0.9439
best	1.0000	0.9588
mean	0.9600	0.9532
spread	0.0881	0.0149
std	0.0221	0.0032

   Cross-validation is 6.8× more stable, and both centre in about the same place — it is not more pessimistic, it is less arbitrary.

The two-minute version. *Eight hundred drives, twenty-nine failures, and one model. Split it five different ways and the AUC comes out 0.961, 0.966, 0.980, 1.000 and 0.990 — every experiment done correctly. Over 200 splits the score runs from 0.885 to a perfect 1.000, and ten of the 200 clear 0.99. Then the expensive part: three models whose honest performance is identical to three decimals each get declared the winner a share of the time, purely by luck of the split. Cross-validation replaces one measurement with five, is nearly seven times more stable across seeds, and comes with a spread you are supposed to report.*

Notebook 02 — `02_bias_variance_and_learning_curves` (self-study on 6 November)

What the error is made of

What it is for. Notebook 1 established that a test score moves. This one asks **why**, and answers with a decomposition that is checked, not asserted — possible only because the true function is ours to choose.

The data. Energy against outdoor temperature, one observation per day, from a curve written for this lesson, $f(t) = 240 + 0.55\,(t-18)^2 + 12\sin(t/4)$: not lesson 3's dataset. 25 days per universe, 300 universes, irreducible noise of **22 kWh**, so a variance of **484** no model can remove.

The concepts, in order.

1. **Three hundred parallel universes.** Three panels — degree 1 wrong in the same way every time (bias) and fanning out too (variance), degree 3 about right, degree 12 flying apart at the edges. In real life you drew one of those lines and never saw the rest.

2. **Measuring the three parts.** First at one temperature, 30 °C, for degree 1: the 300 predictions average 303.6 against a truth of 330.5, so bias² **723.9**, variance **856.1**, noise 484.0, expected error **2,063.9**. Then the same at each of 220 test temperatures, averaged, and the identity holding to floating point:

degree	bias²	variance	noise	total
1	5,250.8	723.8	484.0	6,458.6
2	59.6	70.8	484.0	**614.3**
5	8.9	368.9	484.0	861.8
12	153,537.3	32,174,344.9	484.0	32,328,366.2

   Largest disagreement between the sum and the measured mean squared error (MSE): **2.7** $\times$ **10** $^{\mathbf{-12}}$.

3. **Why degree 1's variance is not the smallest** — the entry the textbook picture does not predict. Each universe redraws its 25 temperatures, and a straight line that cannot follow the curve pivots according to where they fell. Hold the temperatures fixed and the variance column climbs from the first row: **33.9, 55.3, 81.9, 154.5, 709.2, 2,669.8.**

4. **More data moves the answer.** The best degree is 2 at $n = 25$ and **5** at $n = 60$ and $n = 150$: the optimum moves right as the data grows, because there is finally enough of it to pin down the extra coefficients.

5. **Without the truth: training against validation error.** One sample of 25 days, 5-fold cross-validation with `cross_validate(..., return_train_score=True)`, the test set untouched. Degree 1: training **5,155.7**, validation **6,064.3**, both high and close, so bias. Degree 2: lowest validation, **450.4**. Degree 5: training **262.4**, below the noise of 484, validation 615.1, so variance. The same reading as the lab, without knowing the truth; with 150 days degree 5's gap falls from 352.6 to 58.3.

6. **Learning curves**, which add the training-set size. High bias: the curves meet **low**, 0.715 against 0.718, flat from the first point to the last. High variance: training **1.000** against validation 0.847, a gap of 0.153.

7. **And read the slope to the end.** The high-variance curve climbs to **0.857 at 251 examples** and then moves sideways — 0.834, 0.847, 0.835, 0.832, 0.847 — ending *below* its own peak. The last step rises by 0.014, but per round it is −0.030, +0.050, +0.006, −0.009, +0.053: **a slope smaller than its own band is not a slope.**

The two-minute version. *This notebook does something impossible with real data: it draws 300 training sets from a known curve and measures bias, variance and noise separately. The three add up to the measured error to twelve decimal places, so the decomposition is an identity, not a metaphor. Degree 12 has a variance of 32 million. The optimal degree moves from 2 to 5 as the data grows from 25 points to 60, so "the right model" depends on how much data you have. Then the diagnosis without the truth: on one sample of 25 days, training error against validation error*

reads exactly what the lab measured - degree 1 both high (bias), degree 5 training below the noise with a gap (variance). Learning curves add the training-set size: curves that meet low mean bias and more data will not help; a wide gap means variance. And read the slope to the end — ours peaks at 251 examples and then goes flat, which says more rows would buy nothing.

Notebook 03 — `03_leakage_and_honest_search`

Leakage that survives cross-validation

What it is for. Notebook 1 gave them cross-validation. This one shows what it does **not** protect against — because a tool trusted wrongly is more dangerous than no tool.

The data. 2,000 columns of pure noise, 800 rows, 29 positives. The honest AUC of any model built on it is **0.500**.

The concepts, in order.

1. **The catastrophe.** Select the 10 most predictive columns, then cross-validate: **AUC 0.931 ± 0.053**, with four of five folds between 0.93 and 0.98 — which is exactly what a trustworthy result looks like. **Cross-validation did not fail. It was lied to.**
2. **Doing it properly — and the surprise.** Selection inside the pipeline, over 20 CV seeds: mean **0.658**, min 0.576, max 0.759. Better, **and not 0.500**. Searching a large space on a small sample leaves optimism that no rearrangement of the same 800 rows can remove.
3. **Hyperparameter search is the same problem wearing a hat.** 25 combinations on the signal-free table: best candidate **0.7999**, average candidate 0.7265, worst 0.6840. What a practitioner reports is the **maximum of 25 noisy estimates**.
4. **Nested cross-validation.** Reported by the search 0.7999; nested **0.6699 ± 0.1318**; optimism **+0.130** — larger than most differences anyone publishes between methods. Four numbers describing the same signal-free data, each step of discipline costing a chunk of the score.
5. **Doing lesson 4's threshold honestly.** Three-way split: threshold chosen on validation **0.050**, cost on the test set **4,700 €** against **10,540 €** at 0.50. The cheating threshold gives 3,300 € — **and is not reportable**, because it was chosen on the test set.
6. **Seeds.** What to fix and where; and then 30 perfectly reproducible runs scoring from **0.913 to 1.000**. **A fixed seed makes a number repeatable, not correct.**

The two-minute version. *Two thousand columns of noise, eight hundred rows, nothing to learn. Select the ten best features and then cross-validate, and you get 0.93 with a small standard deviation — cross-validation confirms the mistake instead of catching it. Put the selection inside the pipeline and it falls to 0.66, which is better and still not 0.5, because searching a big space on a small sample is optimistic by itself. Then the same thing for hyperparameters: the best of 25 candidates reports 0.80, nested cross-validation says 0.67, so the search was optimistic by thirteen points. And the last section pays lesson 4's debt: choosing the threshold on a validation set instead of the test set costs more on paper and is the only number you are allowed to report.*